Simulating radioactive decay through dice

Aim

To verify the mathematical relation involved in radioactive decay using dice and to determine the half-life of the dice.

Apparatus

Box of dice, Graph paper, Microsoft Excel.

Theory

Natural radioactivity or nuclear decay is a radioactive decay process in which, at a given time, a certain number of atoms disintegrate spontaneously and randomly. In this process, unstable atoms emit ionizing particles or radiations and eventually reach a stable state. There are three types of nuclear decay, viz, Alpha decay, Beta decay and Gamma decay.

In a radioactive decay process, the rate at which atoms undergo decay (dN/dt) is proportional to the number of atoms (N) initially present at a given instant of time which we can write as

$$dN/dt = -\lambda N \qquad ...1$$

where the decay constant λ is related to the probability of decay of an atom of a given radioactive element. One can easily solve the above first order differential equation and obtain the well-known exponential law of radioactive decay as

$$N(t) = N_o e^{-\lambda t} \qquad ...2$$

where N_o is the initial number of atoms present
$N(t)$ is the number of atoms present at a certain time, t; and
λ is a measure of the rate of decay, called the decay constant, which is constant for a given radioactive element.

The decay in Eqn (2) can be further simplified to obtain the decay constant. Taking the natural logarithm of both sides, Eqn (2) becomes

$$\ln N(t) = \ln N_o - \lambda t \qquad \ldots3$$

This equation represents a straight line in $\ln N(t)$ versus t, with slope λ and y-intercept as $\ln (N_o)$. Hence both λ and N_o can be determined. The rate at which a radioactive nuclei decays is measured through its half-life. **The half-life of an element undergoing radioactive decay is the time for the amount to become half of its initial value. Alternatively, we can say that the half-life is the time in which half of the atoms have decayed from their original number.** Half-lives can range from a few microseconds to billions of years. Half-life is very useful such as in carbon dating to find out the age of organic materials.

We can calculate the half-life by substituting $N = N_o/2$ (at $t = T = T_{1/2}$) in Eqn (2)

$$N_o/2 = N_o e^{-\lambda T_{1/2}} \qquad \ldots 4$$

$$½ = e^{-\lambda T_{1/2}}$$

Taking the natural logarithm of both sides, we get the half-life as

$$T_{1/2} = \ln 2/\lambda = 0.693/\lambda$$

The particle radiation resulting from radioactive decay is rather easy to detect experimentally. In fact, their discovery in the 1890s accelerated research in nuclear physics. The radioactive decay processes are of statistical nature. However, the radioactive sources need to be handled with care and need adequate protection. Therefore, an alternate approach to understanding the mathematical nature of the decay process is to simulate it through the rolling of dice and pretend that each die is an atom.

The throwing of dice is a **random event**, the same as the decay of an atom is random. As a result, dice can be used to simulate radioactive decay. To simulate radioactive decay, we need to know when a die has "decayed". The easiest way is to mark one of the faces on a die to represent a "decay" such that when it comes up the die has “decayed” and is removed from the set.

The nuclear decay process can be simulated by the rolling of dice repeatedly, taking out a particular species of dice each time from the lot. In this simulation, $\mathbf{N_0}$ represents the **initial** number of dice taken (100 of similar dice), **N(t)** is the number of dice **remaining** after each round of rolling (decay) and **t** corresponds to the **number of rounds rolled** (equivalent to time).

If a six-faced die is used to simulate the decay process, then the probability of finding one side up (say for example with number-6 or one of the faces marked A) is 1/6. In general, if a die has n faces, then the probability of finding one number up is 1/n.

One can make use of a two-faced coin (probability 1/2), a four-faced die (1/4), or a six-faced (1/6) die to perform this simulation. We have taken six-faced cubical dice for our experiment with a mark A (to indicate decay). The decay constant is $\lambda = p = 1/6$ per rolling of dice. This result is obtained from statistics.

$$T_{1/2} = 0.693/\lambda = 0.693/0.166 = 4.17$$

Procedure

1. About 100 numbers of six-faced cubic dice are taken in a box. The exact number of dice taken is counted. This is the initial number of atoms, $N_o = 100$.

2. The dice are rolled on a large board so that none of them overlaps or interacts (i.e., touch) with each other on the board after rolling. This is the first round of rolling of dice, i.e., r = 1, for which all the dice with the face 'A' are collected and counted. Their number (n) gives the number of decayed atoms.

3. All the dice with the face 'A' are collected, counted and placed in a separate place. Suppose 20 of them have decayed, i.e., the number of dice with the face 'A', n = 20.

   Hence number of dice (atoms) remaining after the first round, $N(t) = 100 - 20 = 80$. This is from experiment.
   Note that t is the number of rolls.
   As given by Eqn (2), the number of atoms remaining is given by $N(t) = N_o e^{-\lambda t} = 100e^{-\lambda 1}$.
   λ for six-faced dice = 0.166
   $N(t) = 100e^{-0.166} = 84.70$
   This shows that theoretically nearly 85 atoms remain after the first round of rolling of dice or 85 atoms remained after a time of t. These numbers are close to the experimentally determined value 80 after the first round of rolling.

4. The number of dice remaining on the board is now collected and rolled again. In the second round of rolling the number of dice with their 'A' face is removed from the board, counted and kept separately. The number of remaining dice is counted and rolled on the board again. The number of dice with the face A and the number of remaining dice are recorded in the table below.

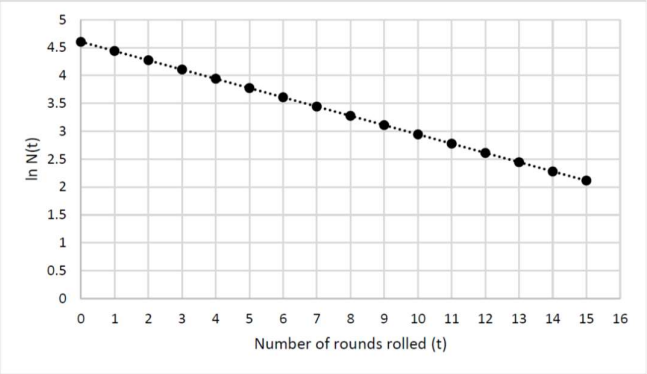
5. Rolling of dice is continued until almost all atoms are decayed. To determine the decay constant experimentally, a graph between the natural logarithm of the experimental values of atoms remaining versus the number of rounds dice have been rolled.

Observations

Fill in the Below table (note that we have given some sample values for reference in the table):

No. of roll (t)	No. of atoms decayed (n)			No. of atoms remaining N(t)		lnN(t) Expt.
	Expt.		Thet.	Expt.	Thet.	
	On Each trial	Total after each trial				
0	0	0	0	100	-	
1	20	20	15	80	85	4.38
2	12	32		68		
3	17	49		51		
4						
5						
6						
7						
8						
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10						
11						
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14						
15						

Plot a graph between ln N(t) and the number of rounds (an example is given below):



Note: This graph is done using theoretical calculations. The actual graph needs to be done using the experiment values.

Slope of the straight line, λ from experiment = …….
Half-life of the dice (atoms) $T_{1/2} = 0.693/\lambda$ = …….

Result

1. We have verified the radioactive decay process of nuclei through the simulation of dice.
2. The decay constant λ is ..... .
3. The half-life is ..... .

Discussion

Discuss the advantages and disadvantages of simulating dice to understand radioactive decay.